Chap 2. Random Variables

The notion of random variable

Sec. 2.1, 2.2

2.1 Discrete Random Variables

- In this chapter and for most of the remainder of the course, we will examine probability models that assign numbers to the outcomes in the sample space.

- When we observe one of these numbers, we refer to the observation as a *random variable*.

- In our notation, the name of a random variable is always a capital letter, for example, X.

- The set of possible values of X is the *range* of X.

- Since we often consider more than one random variable at a time, we denote the range of a random variable by the letter S with a subscript which is the name of the random variable.

- Thus S_X is the range of random variable X, S_Y is the range of random variable Y, and so forth.

Example 2.2

The experiment is to test six integrated circuits and after each test observe whether the circuit is accepted (a) or rejected (r). Each observation is a sequence of six letters where each letter is either a or r. For example, $s_8 =$ aaraaa. The sample space S consists of the 64 possible sequences. A random variable related to this experiment is N, the number of accepted circuits. For outcome s_8, $N = 5$ circuits are accepted. The range of N is $S_N = \{0, 1, \ldots, 6\}$.

Example 2.3

In Example 2.2, the net revenue R obtained for a batch of six integrated circuits is \$5 for each circuit accepted minus \$7 for each circuit rejected. (This is because for each bad circuit that goes out of the factory, it will cost the company \$7 to deal with the customer's complaint and supply a good replacement circuit.) When N circuits are accepted, $6 - N$ circuits are rejected so that the net revenue R is related to N by the function

$$R = g(N) = 5N - 7(6 - N) = 12N - 42 \text{ dollars.} \tag{2.1}$$

Since $S_N = \{0, \ldots, 6\}$, the range of R is

$$S_R = \{-42, -30, -18, -6, 6, 18, 30\}. \tag{2.2}$$

Definition 2.1 Random Variable

A random variable *consists of an experiment with a probability measure $P[\cdot]$ defined on a sample space S and a function that assigns a real number to each outcome in the sample space of the experiment.*

2.1 Notation

- On occasion, it is important to identify the random variable X by the function $X(s)$ that maps the sample outcome s to the corresponding value of the random variable X.

- As needed, we will write $\{X = x\}$ to emphasize that there is a set of sample points $s \in S$ for which $X(s) = x$.

- That is, we have adopted the shorthand notation

$$\{X = x\} = \{s \in S | X(s) = x\}. \tag{2.3}$$

Definition 2.2 Discrete Random Variable

X *is a* discrete *random variable if the range of* X *is a countable set*

$$S_X = \{x_1, x_2, \ldots\}.$$

Section 2.2

Probability Mass Function

Definition 2.4 Probability Mass Function (PMF)

The probability mass function *(PMF) of the discrete random variable* X *is*

$$P_X(x) = P[X = x]$$

Example 2.5

Suppose we observe three calls at a telephone switch where voice calls (v) and data calls (d) are equally likely. Let X denote the number of voice calls, Y the number of data calls, and let $R = XY$. The sample space of the experiment and the corresponding values of the random variables X, Y, and R are

	Outcomes	ddd	ddv	dvd	dvv	vdd	vdv	vvd	vvv
	$P[\cdot]$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$
Random	X	0	1	1	2	1	2	2	3
Variables	Y	3	2	2	1	2	1	1	0
	R	0	2	2	2	2	2	2	0

Example 2.6 Problem

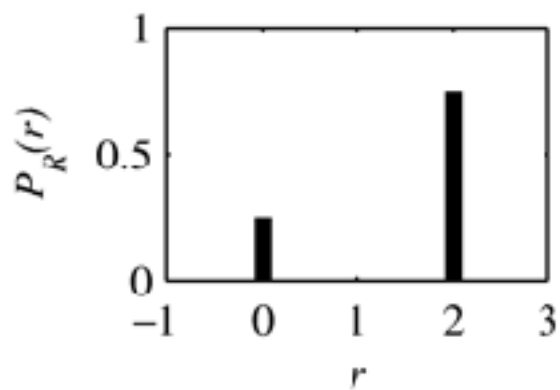
From Example 2.5, what is the PMF of R?

Example 2.6 Solution

From Example 2.5, we see that $R = 0$ if either outcome, DDD or VVV, occurs so that

$$P[R = 0] = P[DDD] + P[VVV] = 1/4. \tag{2.6}$$

For the other six outcomes of the experiment, $R = 2$ so that $P[R = 2] = 6/8$. The PMF of R is



$$P_R(r) = \begin{cases} 1/4 & r = 0, \\ 3/4 & r = 2, \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 2.1

For a discrete random variable X with PMF $P_X(x)$ and range S_X:

(a) For any x, $P_X(x) \geq 0$.

(b) $\sum_{x \in S_X} P_X(x) = 1$.

(c) For any event $B \subset S_X$, the probability that X is in the set B is

$$P[B] = \sum_{x \in B} P_X(x).$$

Families of Discrete Random Variables

Sec. 2.3

2.3 Families of Random Variables

- In practical applications, certain families of random variables appear over and over again in many experiments.

- In each family, the probability mass functions of all the random variables have the same mathematical form.

- They differ only in the values of one or two parameters.

- Depending on the family, the PMF formula contains one or two parameters.

- By assigning numerical values to the parameters, we obtain a specific random variable.

- Our nomenclature for a family consists of the family name followed by one or two parameters in parentheses.

- For example, *binomial* (n, p) refers in general to the family of binomial random variables.

- *Binomial* $(7, 0.1)$ refers to the binomial random variable with parameters $n = 7$ and $p = 0.1$.

Example 2.8

Consider the following experiments:

- Flip a coin and let it land on a table. Observe whether the side facing up is heads or tails. Let X be the number of heads observed.
- Select a student at random and find out her telephone number. Let $X = 0$ if the last digit is even. Otherwise, let $X = 1$.
- Observe one bit transmitted by a modem that is downloading a file from the Internet. Let X be the value of the bit (0 or 1).

All three experiments lead to the probability mass function

$$P_X(x) = \begin{cases} 1/2 & x = 0, \\ 1/2 & x = 1, \\ 0 & \text{otherwise.} \end{cases} \tag{2.13}$$

Definition 2.5 Bernoulli (p) Random Variable

X *is a* Bernoulli (p) *random variable if the PMF of* X *has the form*

$$P_X(x) = \begin{cases} 1-p & x=0 \\ p & x=1 \\ 0 & \textit{otherwise} \end{cases}$$

where the parameter p *is in the range* $0 < p < 1$.

Example 2.9 Problem

Suppose you test one circuit. With probability p, the circuit is rejected. Let X be the probability of a rejected circuit in one test. What is $P_X(x)$?

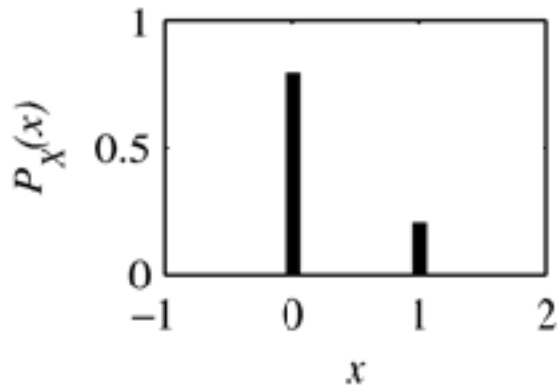
Example 2.9 Solution

Because there are only two outcomes in the sample space, $X = 1$ with probability p and $X = 0$ with probability $1 - p$.

$$P_X(x) = \begin{cases} 1-p & x = 0 \\ p & x = 1 \\ 0 & \text{otherwise} \end{cases} \tag{2.14}$$

Example 2.10

If there is a 0.2 probability of a reject,



$$P_X(x) = \begin{cases} 0.8 & x = 0 \\ 0.2 & x = 1 \\ 0 & \text{otherwise} \end{cases}$$

Example 2.13 Problem

Suppose we test n circuits and each circuit is rejected with probability p independent of the results of other tests. Let K equal the number of rejects in the n tests. Find the PMF $P_K(k)$.

Example 2.13 Solution

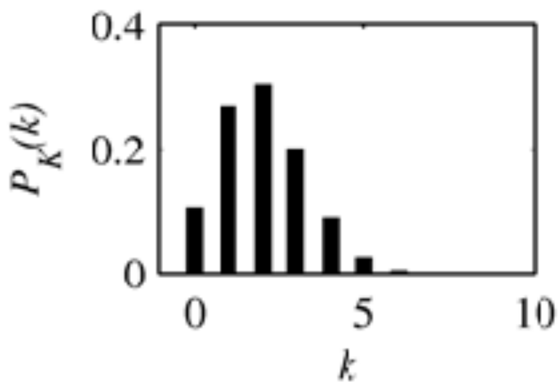
Adopting the vocabulary of Section 1.9, we call each discovery of a defective circuit a *success*, and each test is an independent trial with success probability p. The event $K = k$ corresponds to k successes in n trials, which we have already found, in Equation (1.18), to be the binomial probability

$$P_K(k) = \binom{n}{k} p^k (1-p)^{n-k}. \tag{2.18}$$

K is an example of a *binomial random variable*.

Example 2.14

If there is a 0.2 probability of a reject and we perform 10 tests,



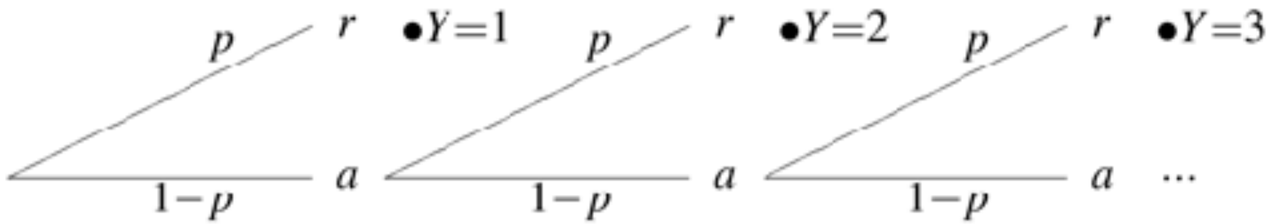
$$P_K(k) = \binom{10}{k}(0.2)^k(0.8)^{10-k}.$$

Example 2.11 Problem

In a test of integrated circuits there is a probability p that each circuit is rejected. Let Y equal the number of tests up to and including the first test that discovers a reject. What is the PMF of Y?

Example 2.11 Solution

The procedure is to keep testing circuits until a reject appears. Using a to denote an accepted circuit and r to denote a reject, the tree is



From the tree, we see that $P[Y = 1] = p$, $P[Y = 2] = p(1 - p)$, $P[Y = 3] = p(1 - p)^2$, and, in general, $P[Y = y] = p(1 - p)^{y-1}$. Therefore,

$$P_Y(y) = \begin{cases} p(1-p)^{y-1} & y = 1, 2, \ldots \\ 0 & \text{otherwise.} \end{cases} \tag{2.16}$$

Y is referred to as a *geometric random variable* because the probabilities in the PMF constitute a geometric series.

Definition 2.6 Geometric (p) Random Variable

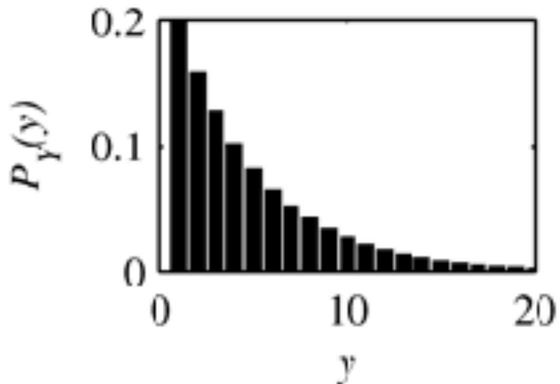
X *is a* geometric (p) *random variable if the PMF of* X *has the form*

$$P_X(x) = \begin{cases} p(1-p)^{x-1} & x = 1, 2, \ldots \\ 0 & \text{otherwise.} \end{cases}$$

where the parameter p *is in the range* $0 < p < 1$.

Example 2.12

If there is a 0.2 probability of a reject,



$$P_Y(y) = \begin{cases} (0.2)(0.8)^{y-1} & y = 1, 2, \ldots \\ 0 & \text{otherwise} \end{cases}$$

Definition 2.8 Pascal (k, p) Random Variable

X *is a* Pascal (k, p) *random variable if the PMF of* X *has the form*

$$P_X(x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}$$

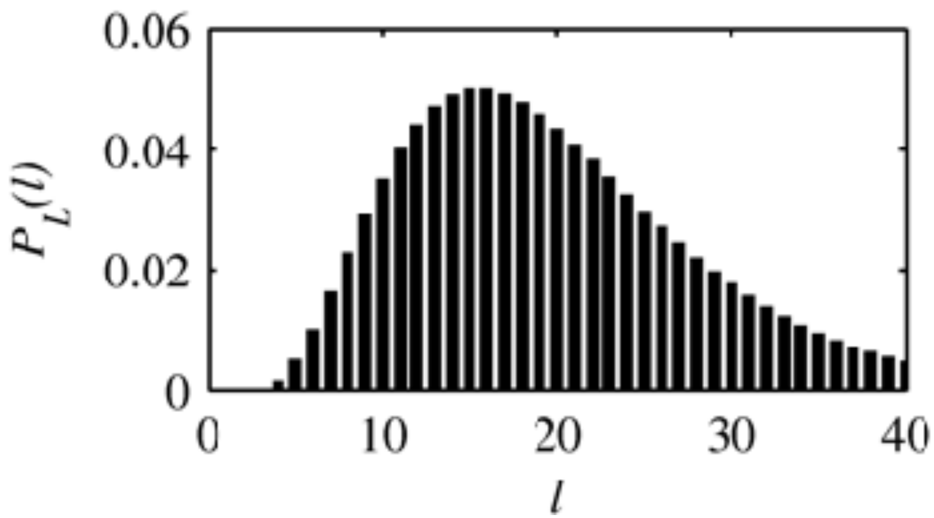
where $0 < p < 1$ *and* k *is an integer such that* $k \geq 1$.

For a sequence of n independent trials with success probability p, a Pascal random variable is the number of trials up to and including the k-th success.

Geometric (p) = Pascal (1,p).

Example 2.16

If there is a 0.2 probability of a reject and we seek four defective circuits, the random variable L is the number of tests necessary to find the four circuits. The PMF is



$$P_L(l) = \binom{l-1}{3}(0.2)^4(0.8)^{l-4}.$$

Summary

- **Bernoulli:** Number of success out of one trial
- **Binomial:** Number of success out of n trials
- **Geometric:** Number of trials until first success
- **Pascal:** Number of trials until success k

Discrete Uniform (k, l) Random

Definition 2.9 Variable

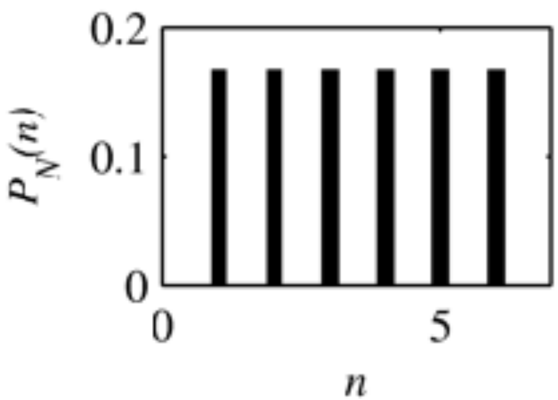
X is a discrete uniform (k, l) random variable if the PMF of X has the form

$$P_X(x) = \begin{cases} 1/(l-k+1) & x = k, k+1, k+2, \ldots, l \\ 0 & \text{otherwise} \end{cases}$$

where the parameters k and l are integers such that $k < l$.

Example 2.18

Roll a fair die. The random variable N is the number of spots that appears on the side facing up. Therefore, N is a discrete uniform $(1, 6)$ random variable and



$$P_N(n) = \begin{cases} 1/6 & n = 1, 2, 3, 4, 5, 6 \\ 0 & \text{otherwise.} \end{cases}$$

Definition 2.10 Poisson (α) Random Variable

X is a Poisson (α) *random variable if the PMF of X has the form*

$$P_X(x) = \begin{cases} \alpha^x e^{-\alpha}/x! & x = 0, 1, 2, \ldots, \\ 0 & \text{otherwise,} \end{cases}$$

where the parameter α is in the range $\alpha > 0$.

Example 2.19 Problem

The number of hits at a Web site in any time interval is a Poisson random variable. A particular site has on average $\lambda = 2$ hits per second. What is the probability that there are no hits in an interval of 0.25 seconds? What is the probability that there are no more than two hits in an interval of one second?

Example 2.19 Solution

In an interval of 0.25 seconds, the number of hits H is a Poisson random variable with $\alpha = \lambda T = (2 \text{ hits/s}) \times (0.25 \text{ s}) = 0.5$ hits. The PMF of H is

$$P_H(h) = \begin{cases} 0.5^h e^{-0.5}/h! & h = 0, 1, 2, \ldots \\ 0 & \text{otherwise.} \end{cases} \tag{2.26}$$

The probability of no hits is

$$P[H = 0] = P_H(0) = (0.5)^0 e^{-0.5}/0! = 0.607. \tag{2.27}$$

In an interval of 1 second, $\alpha = \lambda T = (2 \text{ hits/s}) \times (1\text{s}) = 2$ hits. Letting J denote the number of hits in one second, the PMF of J is

$$P_J(j) = \begin{cases} 2^j e^{-2}/j! & j = 0, 1, 2, \ldots \\ 0 & \text{otherwise.} \end{cases} \tag{2.28}$$

To find the probability of no more than two hits, we note that $\{J \le 2\} = \{J = 0\} \cup \{J = 1\} \cup \{J = 2\}$ is the union of three mutually exclusive events. Therefore,

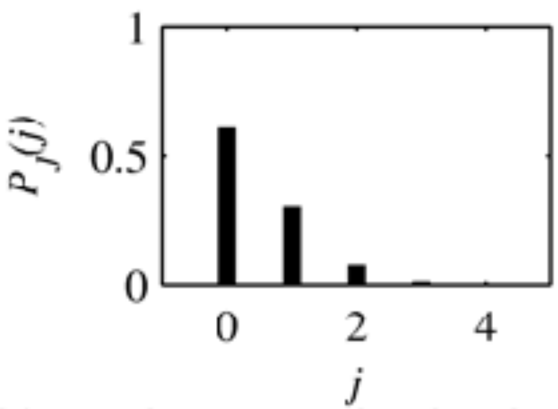
$$P[J \le 2] = P[J = 0] + P[J = 1] + P[J = 2] \tag{2.29}$$

$$= P_J(0) + P_J(1) + P_J(2) \tag{2.30}$$

$$= e^{-2} + 2^1 e^{-2}/1! + 2^2 e^{-2}/2! = 0.677. \tag{2.31}$$

Example 2.21

Calls arrive at random times at a telephone switching office with an average of $\lambda = 0.25$ calls/second. The PMF of the number of calls that arrive in a $T = 2$-second interval is the Poisson (0.5) random variable with PMF



$$P_J(j) = \begin{cases} (0.5)^j e^{-0.5}/j! & j = 0, 1, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

Note that we obtain the same PMF if we define the arrival rate as $\lambda = 60 \cdot 0.25 = 15$ calls per minute and derive the PMF of the number of calls that arrive in $2/60 = 1/30$ minutes.

Cumulative Distribution Function

Sec. 2.4

Definition 2.11 *Cumulative Distribution Function (CDF)*

The cumulative distribution function *(CDF) of random variable X is*

$$F_X(x) = P[X \leq x].$$

Theorem 2.2

For any discrete random variable X with range $S_X = \{x_1, x_2, \ldots\}$ satisfying $x_1 \leq x_2 \leq \ldots$,

(a) $F_X(-\infty) = 0$ and $F_X(\infty) = 1$.

(b) For all $x' \geq x$, $F_X(x') \geq F_X(x)$.

(c) For $x_i \in S_X$ and ϵ, an arbitrarily small positive number,

$$F_X(x_i) - F_X(x_i - \epsilon) = P_X(x_i).$$

(d) $F_X(x) = F_X(x_i)$ for all x such that $x_i \leq x < x_{i+1}$.

2.4 Comment: Theorem 2.2

Each property of Theorem 2.2 has an equivalent statement in words:

(a) Going from left to right on the x-axis, $F_X(x)$ starts at zero and ends at one.

(b) The CDF never decreases as it goes from left to right.

(c) For a discrete random variable X, there is a jump (discontinuity) at each value of $x_i \in S_X$. The height of the jump at x_i is $P_X(x_i)$.

(d) Between jumps, the graph of the CDF of the discrete random variable X is a horizontal line.

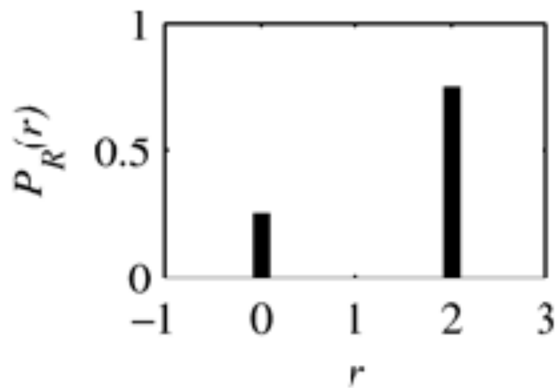
Theorem 2.3

For all $b \geq a$,

$$F_X(b) - F_X(a) = P[a < X \leq b].$$

Example 2.23 Problem

In Example 2.6, we found that random variable R has PMF

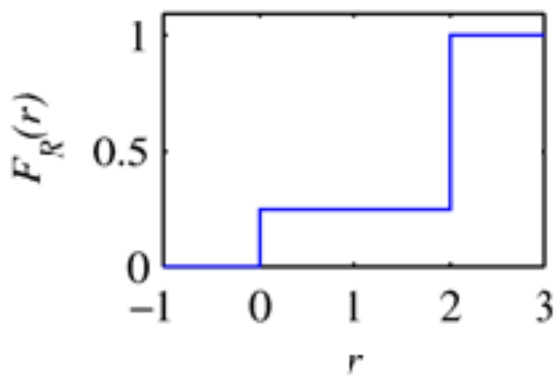


$$P_R(r) = \begin{cases} 1/4 & r = 0, \\ 3/4 & r = 2, \\ 0 & \text{otherwise.} \end{cases}$$

Find and sketch the CDF of random variable R.

Example 2.23 Solution

From the PMF $P_R(r)$, random variable R has CDF



$$F_R(r) = P[R \leq r] = \begin{cases} 0 & r < 0, \\ 1/4 & 0 \leq r < 2, \\ 1 & r \geq 2. \end{cases}$$

Keep in mind that at the discontinuities $r = 0$ and $r = 2$, the values of $F_R(r)$ are the upper values: $F_R(0) = 1/4$, and $F_R(2) = 1$. Math texts call this the *right hand limit* of $F_R(r)$.